

Datacenter Power & Cooling Systems Overview

Reference document describing the power distribution and cooling infrastructure across the facility's server halls.

1. Power Distribution

Utility power enters the facility through two independent feeds from separate substations, providing N+1 redundancy at the facility level. Each feed is rated for 2.5 MW. Power passes through automatic transfer switches (ATS) before reaching the uninterruptible power supply (UPS) systems, which provide ride-through power during the transfer window and during the startup of backup generators. Three diesel generators, each rated at 1.2 MW, provide backup power and can sustain full facility load with one generator offline for maintenance (N+1 generator redundancy).

2. UPS Configuration

UPS units are configured in a 2N topology per server hall, meaning each hall has two fully independent UPS paths, either of which can carry the full hall load alone. Battery runtime at full load is rated at 12 minutes, sufficient to bridge the approximately 45-second generator startup and transfer window with significant margin.

3. Cooling Systems

Cooling is provided by a chilled water system feeding in-row cooling units positioned between server racks. Chilled water is produced by three chillers operating in an N+1 configuration, each rated at 400 tons of cooling capacity. Server halls are arranged in a hot-aisle/cold-aisle containment layout, which improves cooling efficiency by preventing hot and cold air from mixing. Supply air temperature is maintained at 22°C (72°F) with a target humidity range of 40-60% relative humidity.

4. Power Usage Effectiveness (PUE)

PUE is calculated as Total Facility Power divided by IT Equipment Power. The facility's measured PUE over the past 12 months averaged 1.35, meaning for every 1 kW consumed by IT equipment, an additional 0.35 kW is consumed by cooling, power distribution losses, and other overhead. The facility's target PUE, set as part of the annual efficiency goal, is 1.25.

5. Server Hall Capacity Table

Hall	IT Load Capacity (kW)	Current IT Load (kW)	Rack Count	Cooling Type
Hall A	1,800	1,240	60	In-row chilled water
Hall B	1,800	1,610	60	In-row chilled water
Hall C	1,200	540	40	In-row chilled water
Hall D (new)	2,000	180	65	In-row chilled water + rear-door heat exchangers

6. Maintenance Schedule

Generators are load-tested monthly for 30 minutes under simulated facility load. UPS battery health is checked quarterly, with full battery replacement typically required every 5 years. Chiller maintenance follows the manufacturer's schedule of semi-annual service, with the next scheduled service for all three chillers falling in the third week of next month. Hall D's cooling system, being newly commissioned, follows a more frequent monthly inspection schedule for its first year of operation.